

Elasticity

The Arithmetic of Responsiveness

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The core text is N. Gregory Mankiw, *Principles of Microeconomics*.

- **Chapter 5** — Elasticity and Its Application.

Read the chapter alongside these slides.

Lecture outline

1. **Why elasticity** — percentages, because units lie.
2. **Measuring the response of demand** — three ways to one number.
3. **Along the line, and the extremes** — one curve, every elasticity.
4. **What makes demand elastic** — five determinants.
5. **Total revenue** — when a price rise pays and when it backfires.
6. **The elasticity of supply** — the seller's side.
7. **Tax incidence** — who really pays a tax.
8. **Applications** — wheat and the war on drugs.
9. **Income and cross-price elasticity** — two more dials.

PART 1

Why Elasticity

The question supply and demand left open

A demand or supply shift moves price and quantity in a **predictable direction**. The model said nothing about **size**.

- A cafe raises its price 10%. Does revenue rise or fall?
- A cigarette tax: do smokers pay it, or do the companies?
- A record harvest: are farmers richer or poorer?

Every one of these turns on *how much*, not *which way*.

Slopes cannot compare markets

The slope of coffee demand is **10 thousand cups per dollar**. The slope of apartment demand is **units per dollar of rent**.

Which market is more price-sensitive? **The slopes cannot say — their units differ.**

Worse: measure the same coffee in *gallons* (16 cups each) and the slope drops from 10 to 0.625. **Same buyers, same behavior, different number.**

Percentages are unit-free

A **10% change** is a 10% change — in cups, gallons, apartments, dollars, or euros.

So measure responsiveness as a **ratio of percentage changes**:

$$\frac{\% \Delta \text{quantity}}{\% \Delta \text{price}}$$

Units cancel. Markets, goods, countries, decades — all become comparable.

The economist's standard dial

Elasticity. A unit-free measure of how much one variable responds to a change in another, expressed as the ratio of their percentage changes.

The family of dials:

- price elasticity of **demand** — the workhorse;
- price elasticity of **supply**;
- **income** and **cross-price** elasticities, at the end.

Active Learning 1

Two markets in the same city, same month:

- **Coffee:** \$4 $\rightarrow$ \$5; 30,000 $\rightarrow$ 20,000 cups/week.
- **Apartments:** \$2,000 $\rightarrow$ \$2,200 rent; 2,000 $\rightarrow$ 1,950 units/month.

A classmate: *“Apartments barely moved — only 50 units against 10,000 cups — so apartment demand is less responsive.”*

1. Is 50-versus-10,000 a meaningful comparison? Why or why not?
2. Settle it: compute both responsiveness ratios as percentage changes (starting values as the base).

On paper. Three minutes. The *why* is the marks.

PART 2

Measuring the Response of Demand

The price elasticity of demand

Price elasticity of demand (E_d). The percentage change in quantity demanded divided by the percentage change in price:

$$E_d = \frac{\% \Delta Q^d}{\% \Delta P}$$

The law of demand makes the two changes **opposite in sign**, so E_d is negative. By convention we report the absolute value $|E_d|$: **bigger number, more responsive buyers.**

How to read the number

$ E_d $	A 1% price rise cuts quantity demanded by...
1.8	1.8% — quantity moves <i>more</i> than price
1.0	exactly 1%
0.4	only 0.4% — quantity barely moves

Elasticity is a pure number. No cups, no dollars — so any two markets can be compared.

The vocabulary

Condition	Term	Buyers...
$ E_d > 1$	Elastic	respond more than proportionately
$ E_d = 1$	Unit elastic	respond exactly in proportion
$ E_d < 1$	Inelastic	respond less than proportionately
$ E_d = 0$	Perfectly inelastic	do not respond at all
$ E_d \rightarrow \infty$	Perfectly elastic	flee any price rise entirely

The boundary is **one**: quantity moving faster or slower than price.

A problem: the answer depends on the direction

Two points on the coffee curve $Q^d = 70 - 10P$: $A = (\$4, 30)$ and $B = (\$5, 20)$.

Walking...	% ΔQ	% ΔP	ratio
$A \rightarrow B$	$-10/30 = -33.3\%$	$+1/4 = +25\%$	1.33
$B \rightarrow A$	$+10/20 = +50\%$	$-1/5 = -20\%$	2.50

Same two points, two elasticities. The base of each percentage changed.

Fix 1 — the midpoint method

Use the **average** of the two endpoints as the base, in both numerator and denominator:

$$|E_d| = \left| \frac{(Q_2 - Q_1) / \left[\frac{Q_1 + Q_2}{2} \right]}{(P_2 - P_1) / \left[\frac{P_1 + P_2}{2} \right]} \right|$$

For $A = (4, 30)$, $B = (5, 20)$:

$$\frac{10/25}{1/4.5} = \frac{40.0\%}{22.2\%} = \mathbf{1.80} \quad \text{— either direction.}$$

Use it whenever the data are two observed points.

Fix 2 — the slope method, when you have the equation

From $Q^d = 70 - 10P$: the coefficient **10** says each \$1 of price moves quantity by 10 (thousand cups). That is the slope — **read it straight off the equation**.

Scale it into percentage terms:

$$|E_d| = \underbrace{10}_{\text{slope}} \times \frac{P}{Q}$$

At the point $(P, Q) = (4.50, 25)$: $|E_d| = 10 \times \frac{4.50}{25} = \mathbf{1.80}$.

Two routes, one answer

The point $(4.50, 25)$ is the **center of the arc** from A to B .

Midpoint method across the arc $A \rightarrow B$	1.80
Slope method at the arc's center	1.80

Two independent routes agree. If yours ever disagree, one of them is wrong — go looking.

Active Learning 2

The campus smoothie bar's demand, two observations: at $P = \$3$, $Q = 900/\text{week}$; at $P = \$5$, $Q = 700/\text{week}$.

1. Compute the ratio of percentage changes walking $\$3 \rightarrow \5 , then walking $\$5 \rightarrow \3 . (They will differ — that is the point.)
2. Compute the midpoint-method elasticity.
3. **Check it:** the slope here is 100 smoothies per dollar. Apply the slope method at the arc's center ($\$4, 800$). Do the two routes agree?

On paper. Four minutes. Show every substitution.

PART 3

Along the Line, and the Extremes

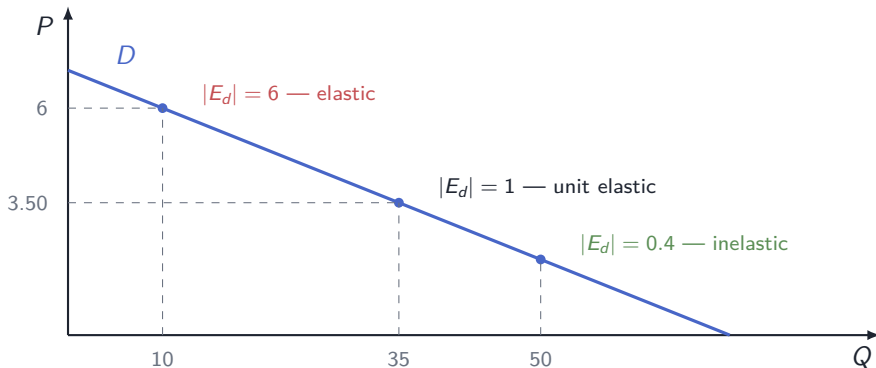
Plug the numbers in — elasticity along the coffee curve

$Q^d = 70 - 10P$, slope 10 everywhere. Yet:

P (\$)	Q (000s)	Substitute $10 \times P/Q$	$ E_d $	Verdict
6	10	$10 \times 6/10$	6.00	elastic
5	20	$10 \times 5/20$	2.50	elastic
4	30	$10 \times 4/30$	1.33	elastic
3.50	35	$10 \times 3.5/35$	1.00	unit elastic
3	40	$10 \times 3/40$	0.75	inelastic
2	50	$10 \times 2/50$	0.40	inelastic
1	60	$10 \times 1/60$	0.17	inelastic

One straight line. Seven different elasticities.

Draw the graph



Elastic at the top, unit elastic at the midpoint, inelastic at the bottom.

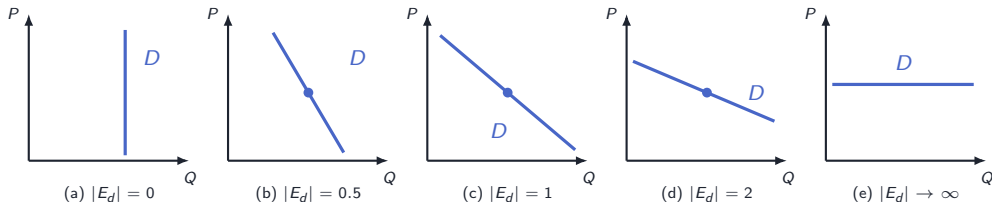
Slope is not elasticity

Slope — units per dollar. Constant along a straight line.

Elasticity — percentage per percentage. Changes along the same line, because P/Q changes.

“The curve is a straight line, so elasticity is constant” — the classic exam error. You now know why it is wrong.

The whole range of demand curves



From no response at all to total flight. **Flatter means more elastic** — the three middle panels are measured at the same marked point.

The two extremes are real

Perfectly inelastic ($|E_d| = 0$, vertical). A transplant patient's anti-rejection medication: the prescribed dose, at almost any price. **Price changes move no quantity at all.**

Perfectly elastic ($|E_d| \rightarrow \infty$, horizontal). One wheat farmer among thousands: charge a cent above the market and sell **nothing**.

The horizontal case is the demand curve facing a single seller in a market with thousands of identical rivals.

Active Learning 3

A demand curve: $Q^d = 40 - 4P$.

1. **Predict before computing:** at what price is this curve unit elastic? (You know where unit elasticity sits on a straight line — use that, no arithmetic.)
2. Compute $|E_d|$ at $P = 8$, at $P = 5$, and at $P = 2$, showing the substitution each time.
3. Do the three numbers confirm your prediction and the top-to-bottom pattern?

On paper. Four minutes.

PART 4

What Makes Demand Elastic

Substitutes, and how you draw the market

1. Close substitutes. Butter's price rises; buyers switch to margarine. Eggs have no stand-in, so egg buyers stay. **More substitutes → more elastic.**

2. Market definition. *Food* has no substitute at all — demand inelastic. *One bakery's sourdough loaf* has a substitute in every other bakery — demand highly elastic. **Narrower market → more elastic.**

3. Necessities vs. luxuries. A doctor's visit happens at almost any price; a sailboat does not. **Luxury** → **more elastic** — and which is which lives in the buyer's preferences, not the good.

4. Share of income. Rent doubling rearranges your life; salt doubling goes unnoticed. **Bigger budget share** → **more elastic**.

5. Time horizon. Gasoline jumps today: this month you still drive to work. Over years — a smaller car, a carpool, a move closer to work.

Demand is more elastic in the long run: substitution takes time.

Short-run and long-run elasticity can differ severalfold. **Any policy priced off the short-run number will misfire in the long run.**

The determinants, measured

Good	Estimated $ E_d $
Eggs	0.1
Healthcare	0.2
Cigarettes	0.4
Rice	0.5
Housing	0.7
Beef	1.6
Restaurant meals	2.3

Necessities at the top, discretionary goods at the bottom. **Treat any single estimate with caution.**

Active Learning 4

Rank these three by $|E_d|$, **smallest to largest**:

tap water

beef

meals at one particular restaurant

1. Write the ranking.
2. Justify each position using the determinants — *name the determinant doing the work*.
3. Which single determinant does this ranking turn on most?

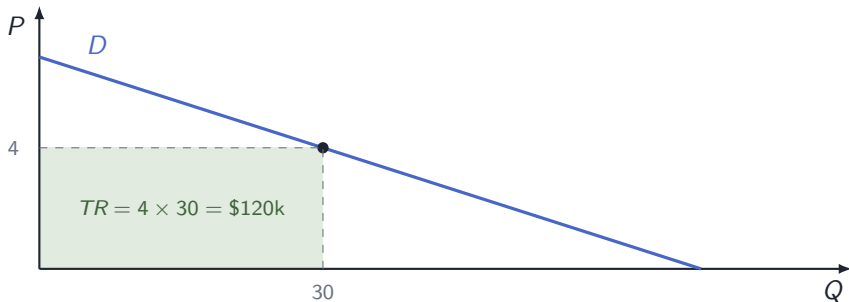
On paper. Three minutes. No numbers to crunch — the reasoning is the marks.

PART 5

Total Revenue

Total revenue is a rectangle

Total revenue (TR). The total amount buyers pay and sellers receive: $TR = P \times Q$.



Height P , width Q . **What happens to the rectangle when the price changes?**

Two effects fight

Raise the price and:

- the **price effect** — each unit sells for more — pushes *TR* **up**;
- the **quantity effect** — fewer units sell — pushes *TR* **down**.

	Price rises	Price falls
Elastic ($ E_d > 1$)	<i>TR</i> falls	<i>TR</i> rises
Unit elastic ($ E_d = 1$)	<i>TR</i> unchanged	<i>TR</i> unchanged
Inelastic ($ E_d < 1$)	<i>TR</i> rises	<i>TR</i> falls

Elastic: quantity effect wins. Inelastic: price effect wins.

Watch elastic demand punish a price rise

$|E_d| = 2$. Start: 100 units at \$10 $\Rightarrow TR_1 = \$1,000$.

Raise the price 20% to \$12:

$$\% \Delta Q = 2 \times 20\% = 40\% \text{ fall} \Rightarrow Q = 100 - 40 = 60$$

$$TR_2 = 12 \times 60 = \$720$$

Revenue fell \$280. The 40 lost units overwhelmed the extra \$2 on the 60 that remained.

Watch inelastic demand reward the same move

$|E_d| = 0.5$. Same start: 100 units at \$10 $\Rightarrow TR_1 = \$1,000$.

Raise the price 20% to \$12:

$$\% \Delta Q = 0.5 \times 20\% = 10\% \text{ fall} \Rightarrow Q = 100 - 10 = 90$$

$$TR_2 = 12 \times 90 = \$1,080$$

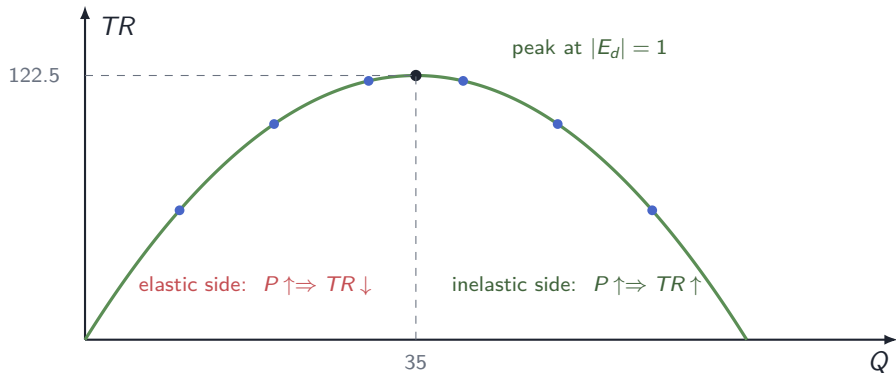
Same price move, opposite outcome. The only thing that changed was the elasticity.

Plug the numbers in — revenue along the coffee curve

P (\$)	Q (000s)	$TR = P \times Q$ (\$000s)	$ E_d $	Region
1	60	$1 \times 60 = 60.0$	0.17	inelastic
2	50	$2 \times 50 = 100.0$	0.40	inelastic
3	40	$3 \times 40 = 120.0$	0.75	inelastic
3.50	35	$3.5 \times 35 = 122.5$	1.00	unit elastic
4	30	$4 \times 30 = 120.0$	1.33	elastic
5	20	$5 \times 20 = 100.0$	2.50	elastic
6	10	$6 \times 10 = 60.0$	6.00	elastic

Climbing through the inelastic rows, TR **rises**. Past \$3.50, into the elastic rows, it **falls**. The peak sits exactly at $|E_d| = 1$.

Draw the graph — the revenue hill



Dots from the table, curve from $TR = 7Q - Q^2/10$ — coinciding.

The rule to carry

Inelastic demand: raise the price, revenue rises.

Elastic demand: raise the price, revenue falls.

Revenue peaks where $|E_d| = 1$.

This one rule runs every revenue argument in economics — farm incomes, airline pricing, the choices facing any seller with market power.

Extensions — the same rule by calculus

Extension material — not needed for anything else here. The result above was already established numerically; this is the same result in one line, for those who have met derivatives.

Differentiate $TR = P \times Q$ along the demand curve:

$$\frac{dTR}{dP} = P \frac{dQ}{dP} + Q = -|E_d| Q + Q = Q(1 - |E_d|)$$

Positive when $|E_d| < 1$, zero at $|E_d| = 1$, negative when $|E_d| > 1$ — **the table, the hill, and the rule, in one line.**

Active Learning 5

Two organizations, each pondering a **25% price rise**:

- **City bus system:** fare \$2.00, ridership 10,000/day, $|E_d| = 0.4$.
- **Stadium concert:** ticket \$40, sales 4,000, $|E_d| = 2$.

1. **Predict first**, from the elasticities alone: whose revenue rises, whose falls?
2. Now compute both new quantities and both new revenues. Show the substitutions.
3. One sentence: what did the two answers turn on?

On paper. Four minutes.

PART 6

The Elasticity of Supply

Price elasticity of supply (E_s). The percentage change in quantity supplied divided by the percentage change in price:

$$E_s = \frac{\% \Delta Q^s}{\% \Delta P}$$

Supply slopes **up**, so E_s is **positive** — no absolute value needed. The vocabulary transfers whole: elastic (> 1), unit ($= 1$), inelastic (< 1), and the two perfect extremes.

Computing it — same midpoint method

A bakery's sourdough supply: at $P = \$3$ a loaf, 3,000 loaves/week; at $P = \$4$, 5,000.

$$\% \Delta Q = \frac{2,000}{4,000} = 50\% \quad \% \Delta P = \frac{1}{3.50} = 28.6\%$$

$$E_s = \frac{50\%}{28.6\%} = \mathbf{1.75} \quad \text{— elastic.}$$

Ovens run longer, an extra baker comes in: **quantity can chase the price.**

What lets quantity chase the price

1. Production flexibility. A print shop can add a shift within a day. A vineyard cannot add a harvest — the vines are planted.

2. Input availability. Expansion needs workers, materials, machines — **at hand**. If skilled labor is scarce, willing firms still cannot grow.

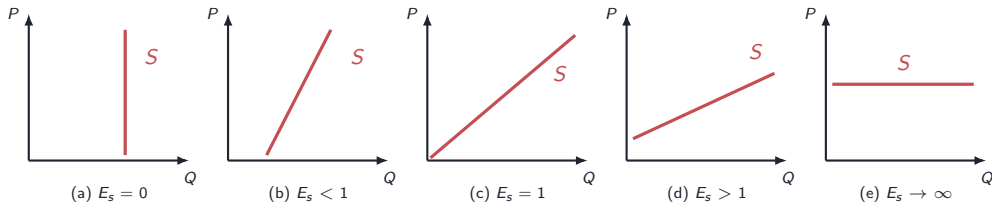
The hard limit: some quantities are **fixed by nature**. Beachfront land — no price mints more coastline. $E_s = 0$.

3. Time horizon. The strongest determinant, exactly as on the demand side:

- **Short run** — capacity is fixed; output can only stretch within it. Supply **less** elastic.
- **Long run** — firms build plants, and **new firms enter**. Supply **more** elastic.

Almost every supply curve is more elastic in the long run.

The whole range of supply curves



Vertical: beachfront land. Horizontal: an industry that can supply any amount at the going price, and nothing below it.

Active Learning 6

1. **Classify first, no computing:** which has the more elastic supply, and why — **(a)** oceanfront building lots in a beach town, or **(b)** custom-printed t-shirts?
2. Now compute: a print shop supplies 200 shirts/day at $P = \$10$ and 260/day at $P = \$12$. Find E_s by the midpoint method, showing the substitution, and classify it.

On paper. Three minutes. Name the determinant in part 1.

PART 7

Tax Incidence: Who Really Pays

The check-writer is not the payer

A city puts a **\$1 tax on every pizza sold, collected from the seller.**

Sellers will try to pass it on as a higher price. **Can they? How much of it?**

The law decides who mails the check. Elasticity decides who bears the burden.

Plan: **one market, one \$1 tax, two elasticity configurations.**

Configuration A — write the equations, find equilibrium

Buyers hooked, sellers flexible:

$$Q^d = 100 - 5P \qquad Q^s = 20P - 50$$

$$100 - 5P = 20P - 50 \Rightarrow 150 = 25P \Rightarrow P^* = \$6, \quad Q^* = 100 - 5(6) = 70$$

Elasticities at the equilibrium (slope method):

$$|E_d| = 5 \times \frac{6}{70} = \mathbf{0.43} \qquad E_s = 20 \times \frac{6}{70} = \mathbf{1.71}$$

Demand inelastic, supply elastic. Prediction before solving: who bears more?

Configuration A — impose the \$1 tax

Sellers now keep $P - 1$, so supply becomes

$$Q^s = 20(P - 1) - 50 = 20P - 70$$

Solving $100 - 5P = 20P - 70$ gives $P = \$6.80$.

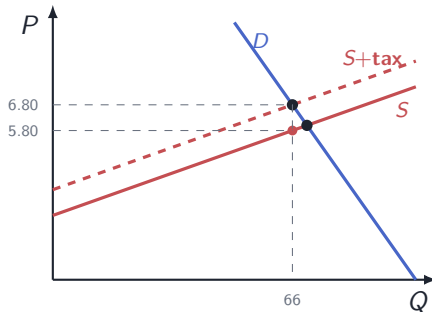
Buyers now pay	\$6.80
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up from \$6.00	+\$0.80
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Sellers now keep	\$5.80
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down from \$6.00	-\$0.20
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Check: $0.80 + 0.20$	$= \$1.00$	✓
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Buyers bear 80 cents of every dollar. The inelastic side paid.

Configuration B — swap the elasticities, repeat everything

Same equilibrium, roles reversed — buyers flexible, sellers stuck:

$$Q^d = 190 - 20P \quad Q^s = 5P + 40 \quad \Rightarrow \quad P^* = \$6, Q^* = 70$$

$$|E_d| = 20 \times \frac{6}{70} = \mathbf{1.71} \quad E_s = 5 \times \frac{6}{70} = \mathbf{0.43}$$

Same \$1 tax: $Q^s = 5(P - 1) + 40 \Rightarrow 190 - 20P = 5P + 35 \Rightarrow P = \6.20

Buyers pay \$6.20	+\$0.20
Sellers keep \$5.20	-\$0.80
Check: $0.20 + 0.80 = \$1.00$	✓

Same equilibrium, same tax — the burden flipped.

The incidence rule

The more inelastic side of the market bears more of the tax.

	$ E_d $	E_s	Buyers bear	Sellers bear
Config. A	0.43	1.71	\$0.80	\$0.20
Config. B	1.71	0.43	\$0.20	\$0.80

Inelastic means **nowhere to run**: buyers who won't cut back, or sellers who can't cut production, absorb the price move. **Who mails the check never mattered.**

Half a story — deliberately

The **who**: relative elasticities split the burden.

The other half, which belongs with the study of government policy:

- the **tax wedge** on the diagram,
- **revenue** the government collects,
- the trades the tax **prevents** — note Q fell $70 \rightarrow 66$ in *both* configurations.

Neither half repeats the other. Together they are the tax analysis.

Active Learning 7

Food-truck lunches (hundreds/day, P in \$):

$$Q^d = 60 - 2P \qquad Q^s = 8P - 40$$

1. Solve for the no-tax equilibrium.
2. The city imposes a **\$2 per lunch tax, collected from the trucks. Predict before solving:** who bears more, and why? (Compute both elasticities at the equilibrium — slope method.)
3. Now solve with the tax: buyers' price, sellers' take, and each side's burden.
Check: the two burdens must sum to \$2.00.

On paper. Five minutes. Prediction committed before part 3.

PART 8

Applications

Good news arrives in the wheat market

A better wheat variety raises yields. **Supply shifts right; demand — a staple, few substitutes — stays put and is inelastic.**

The market, before:

$$Q^d = 130 - 10P \quad Q^s = 50P - 50$$

$$130 - 10P = 50P - 50 \Rightarrow P^* = \$3, \quad Q^* = 100 \quad TR = \$300$$

Predict: bigger harvests — are farmers richer or poorer?

The same market, after

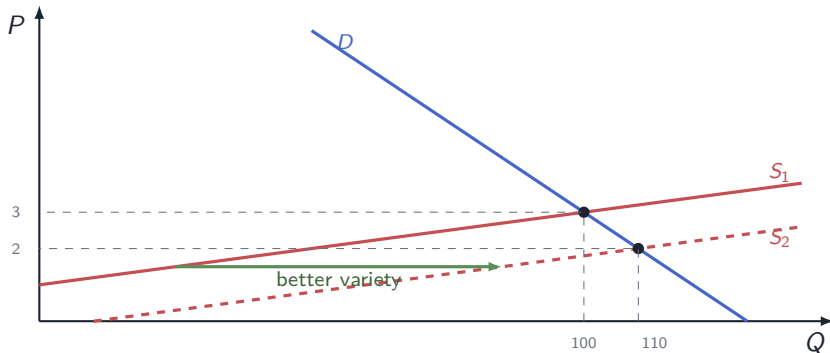
New supply: $Q^s = 50P + 10$. Demand unchanged.

$$130 - 10P = 50P + 10 \Rightarrow 120 = 60P \Rightarrow P^* = \$2, \quad Q^* = 110$$

	P	Q	$TR = P \times Q$
Before	\$3	100	\$300
After	\$2	110	\$220
Change	-33%	+10%	-\$80

More wheat, less revenue. Inelastic demand: the price fell three times as fast as quantity rose.

Draw the graph



A steep D met by a rightward S shift: P falls a lot, Q rises a little.

This paradox emptied the countryside

In 1900, about **40% of the American labor force farmed**. By 2020, **1.4%**.

Two centuries of better seed, machinery, fertilizer — supply marching right against **inelastic food demand**. Prices fell, farm revenues fell, and labor moved to town.

Consumers captured the gains from farm technology. Farmers, as a group, did not.

The same logic, darker: the war on drugs

Interdiction — seize supply at the border. S shifts left. Addiction makes demand **inelastic**: price soars, quantity barely falls, so **total spending** $P \times Q$ **rises** — and much of it is financed by theft. **More drug-related crime, not less.**

Education — shift *demand* left instead. Price falls *and* quantity falls: **spending falls unambiguously.**

Same goal, opposite predictions — separated only by which curve the policy moves.

Active Learning 8

A newspaper headline, now with numbers: *a late frost ruins part of the orange crop.*

Before: $P = \$2/\text{lb}$, $Q = 300$ (millions of lbs), so $TR = \$600\text{m}$.

After the frost: $P = \$3.20$, $Q = 220$.

1. **Predict first:** growers' total revenue — up or down? What single fact about orange demand decides it?
2. Compute the new TR , and the midpoint elasticity between the two points. Do they agree with your prediction?
3. Wheat farmers' *good* harvest made them poorer; orange growers' *bad* harvest made them richer. Reconcile in one sentence.

On paper. Four minutes.

PART 9

Income and Cross-Price Elasticity

Income elasticity of demand (E_I). The percentage change in quantity demanded divided by the percentage change in income:

$$E_I = \frac{\% \Delta Q^d}{\% \Delta I}$$

The sign carries the information — so here, keep it. Incomes rise 10%; store-brand instant noodles fall 3%:

$$E_I = \frac{-3\%}{+10\%} = -0.3$$

What the sign and size say

E_I	Term	Example
$E_I < 0$	Inferior	store-brand noodles, bus rides
$0 < E_I < 1$	Necessity	groceries as a whole
$E_I > 1$	Luxury	foreign vacations

Inferior goods are **fled** as income rises; necessities grow **slower** than income; luxuries grow **faster**.

Same deal again: the other good's price

Cross-price elasticity (E_{XY}). The percentage change in the quantity of good X demanded, divided by the percentage change in the *price of good Y* :

$$E_{XY} = \frac{\% \Delta Q_X^d}{\% \Delta P_Y}$$

$E_{XY} > 0$	Substitutes	tea $\uparrow$ when coffee price rises
$E_{XY} < 0$	Complements	ink $\downarrow$ when printer price rises

Reading a cross-price number

Gasoline prices rise **20%**; the quantity of new cars demanded falls **5%**:

$$E_{XY} = \frac{-5\%}{+20\%} = -0.25$$

Negative $\Rightarrow$ **complements** — cars are useless without fuel. Small (0.25) $\Rightarrow$ the link is real but **loose**.

Sign gives the relationship; size gives its strength.

Active Learning 9

Quick fire — classify each, with the computed elasticity as your evidence:

1. Incomes rise 5%; streaming subscriptions rise 8%.
2. Incomes rise 5%; bus rides fall 2%.
3. Coffee prices rise 10%; tea sales rise 4%.
4. Coffee prices rise 10%; coffee-creamers sales fall 6%.

On paper. Two minutes. Number first, then the word it justifies.

How much matters as much as which way.

Supply and demand give the direction; elasticity — a pure ratio of percentage changes — gives the size. And the size decides the things people actually argue about: whether a price rise pays, who really bears a tax, and why a good harvest can be bad news.

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